

WarpTalk

Speak once.
Be understood everywhere.

An AI Speech Translation Platform for
Real-Time Multilingual
Communication with Voice Cloning.

PROJECT FOCUS

Communication without
language barriers.

PRESENTED BY

GSU26SE16

SUPERVISOR

Ms. Thân Thị Ngọc Vân

COVER
01

THE PEOPLE BEHIND WARPTALK

Team members

Four students building **real-time multilingual communication** through AI speech translation and voice cloning.

TEAM LEADER

Huynh
Thai Tu

SE183307

01

MEMBER

Ngo Xuan
Hanh Nhi

SE192260

02

MEMBER

Huynh
Ngoc Ky

SE182995

03

MEMBER

Tran Manh
Tuan

SE180360

04

PRESENTATION STRUCTURE

Table of contents

From the communication problem to a **validated system** built for real-time multilingual conversations.

SECTION I

Context & Problems

Why language barriers still disrupt live communication.

01

SECTION II

Proposed Solution

How WarpTalk translates speech while preserving voice identity.

02

SECTION III

System Architecture

The services and data flow behind the platform.

03

SECTION IV

Tech Stack

Technologies selected for scale, latency, and reliability.

04

SECTION V

Functional Requirements

The capabilities required across each user role.

05

SECTION VI

Core Workflows

How users move through the system's primary flows.

06

SECTION VII

Project Achievements

Delivery outcomes, validation, and lessons learned.

07

SECTION VIII

Questions & Answers

Discussion with the review committee.

08

SECTION 01 OF 08

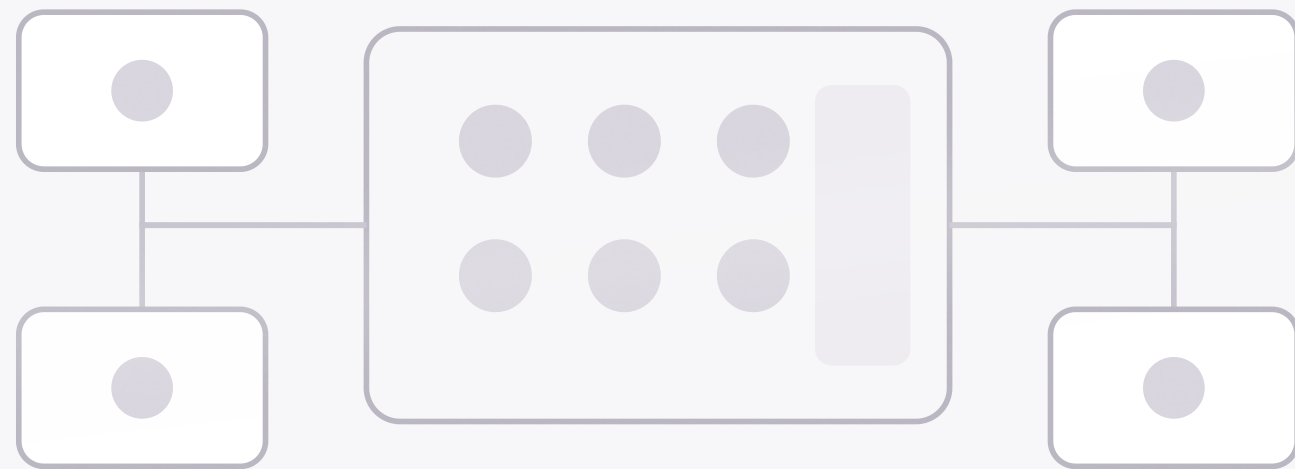
CONTEXT & PROBLEMS

Why language barriers still disrupt live communication.

01

SECTION I · CONTEXT

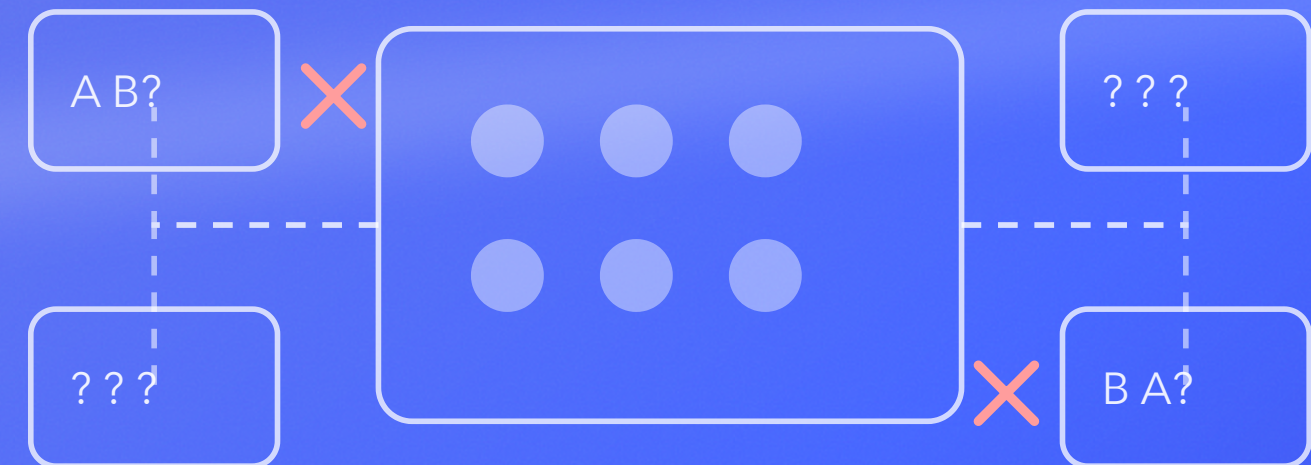
Meetings went global. Understanding did not.



Global teams meet online every day.

Video calls are now the default room. The people in them no longer share a country, a timezone, or a first language.

Distributed teams already meet online every day – the **language gap** is what still stands between a conversation and a decision.

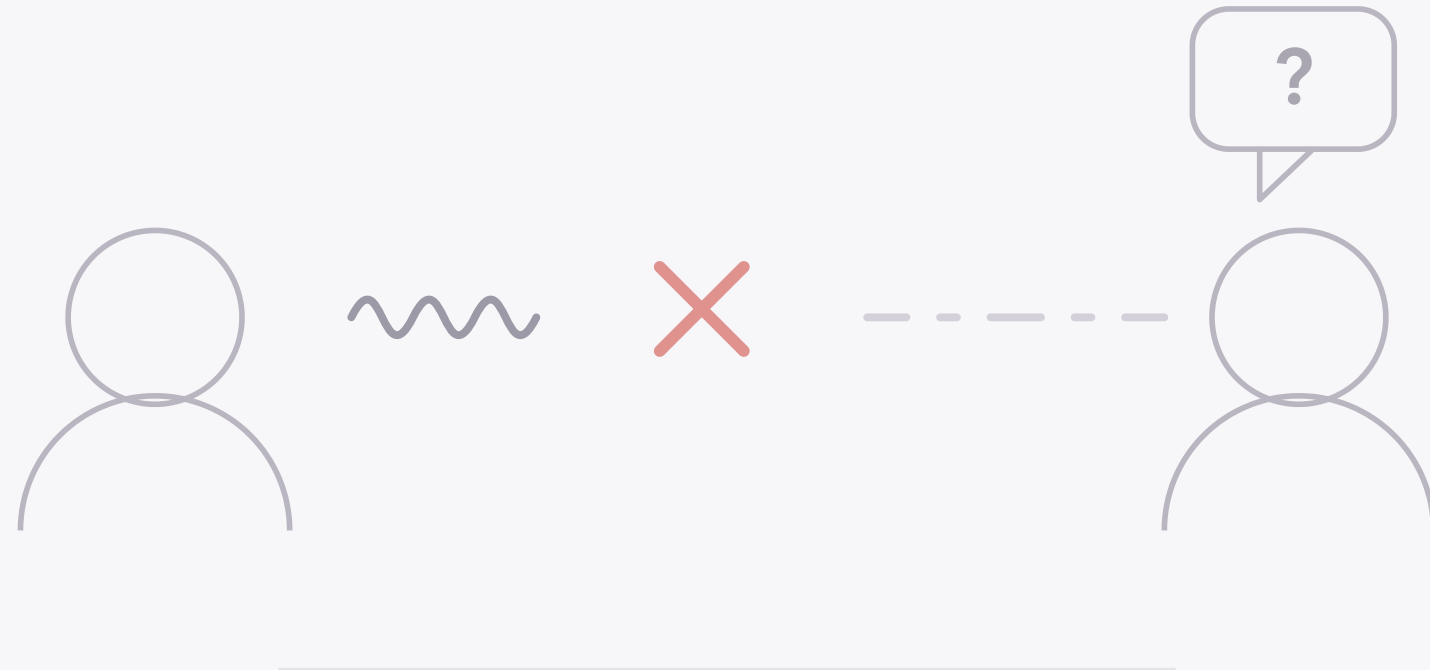


Language barriers still slow decisions down.

Meaning is lost in the gap between what is said and what is understood – so the meeting ends without the decision it was called for.

SECTION I · PROBLEMS

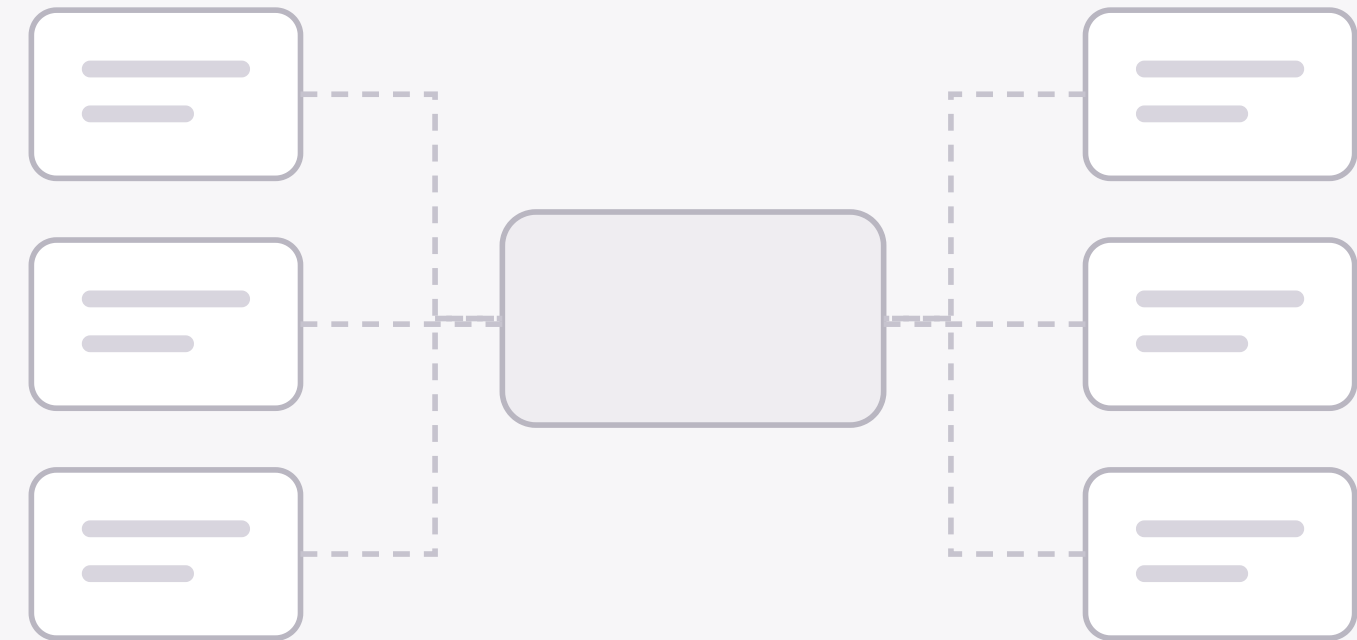
Two problems, one broken conversation.



Language differences cause communication problems.

Subtitles arrive late, interpreters do not scale, and the speaker's own voice – the part people actually trust – disappears entirely.

Teams pay twice: once for the **language gap**, and again for the pile of tools bolted on to close it.



Multiple tools make collaboration more complicated.

Meeting, captioning, translation, notes and storage each live in a different product – so the record of the conversation is scattered before anyone can use it.

SECTION 02 OF 08

PROPOSED SOLUTION

How WarpTalk translates speech while preserving voice identity.

02

SECTION II · PROPOSED SOLUTION

A B2B platform, six key features.

WarpTalk lets an organization join the system as a **workspace** and run multilingual meetings end to end – from live speech to a searchable record.

1

Real-time Multilingual Translation Meeting

Everyone speaks their own language; everyone hears the room in theirs, live.

2

Voice Cloning

The translation is delivered in the speaker's own voice, with consent and a safe fallback.

3

Live Transcript & WarpBot Assistant

The conversation is transcribed as it happens and answerable by an in-product AI assistant.

4

Google Meet Integration

Teams keep the meeting platform they already use – WarpTalk joins it as a bridge.

5

AI Summaries & Suggestions

Overview, decisions and action items are generated the moment the meeting ends.

6

Workspace Collaboration & AI Credits

Subscription plans, shared AI credits and usage tracking make it operable by an organization.

SECTION 03 OF 08

SYSTEM ARCHITECTURE

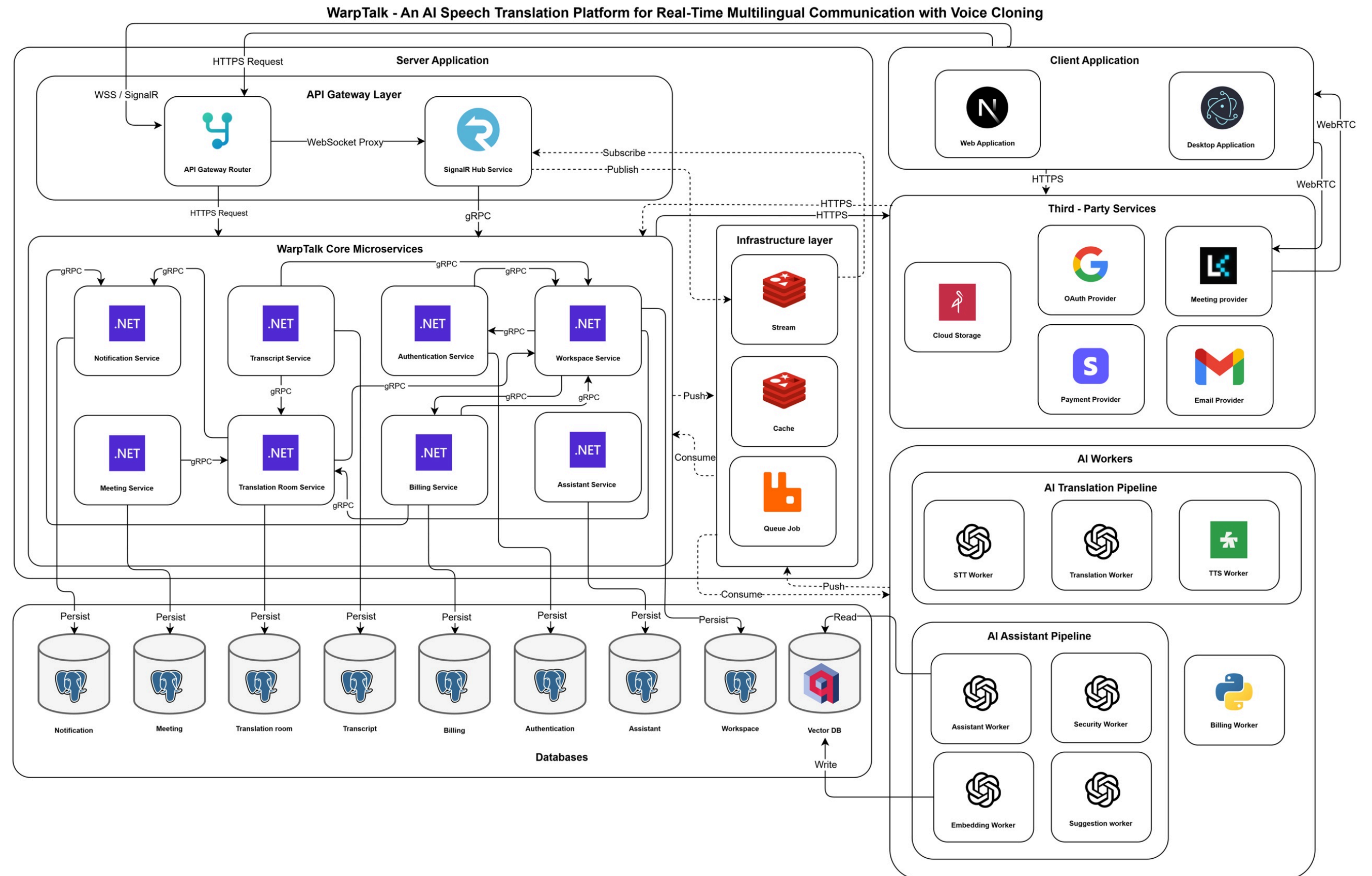
The services and data flow behind the platform.

03

SECTION III · SYSTEM ARCHITECTURE

How the pieces fit

One gateway in front, **a database per service** behind, and two AI pipelines that reach the services only through the stream.



SECTION 04 OF 08

TECH STACK

Technologies selected for latency, reliability and scale.

04

SECTION IV · TECH STACK

Six layers, one system.

A .NET core platform and a Python AI tier over two databases – plus the parts we **deliberately did not build**.

01 Backend & AI

 .NET

 .NET Core

 Python

 FastAPI

 Redis

 RabbitMQ

 SignalR

 YARP

02 Frontend

 ReactJS

 Next.js

 ElectronJS

 TailwindCSS

03 Database

 PostgreSQL

 Qdrant

04 Third-party services

 OpenAI

 Cartesia

 LiveKit

 Google

 Stripe

 Cloudflare

 Resend

05 Infrastructure & delivery

 GitHub

 Docker

 Terraform

 K3s

 Vietnix

06 Observability

 Prometheus

 Grafana

SECTION 05 OF 08

FUNCTIONAL REQUIREMENTS

Capabilities required across each user role.

05

SECTION V · FUNCTIONAL REQUIREMENTS

Four actors, four surfaces.

Every requirement in the system belongs to exactly one of these roles – and each role gets **its own screen**.

ROLE I



Participants

Anyone in the room. Joins, picks a language, listens and reads.

01

ROLE II



Host

Owens the meeting. Creates it, runs translation, and signs off the record.

02

ROLE III



Workspace Owner

Owens the organization. Buys the plan, sets policy, adds people and knowledge.

03

ROLE IV



WarpTalk Administrator

Owens the platform. Manages every workspace, plan and audit trail.

04

ROLE I OF IV

Participants

The lightest role in the product: **join, choose a language, be understood.** Nothing else is required of them.

- | | | |
|----|--|--|
| 01 | Join meetings | Open an invite or a room code and enter – as a workspace member or an external guest. |
| 02 | Set device & language | Pick microphone, speaker and the language they will speak and hear before joining. |
| 03 | Use real-time translation | Hear the room in their own language, in the speaker's cloned voice, with live captions. |
| 04 | View transcript & artifacts | Read the running transcript during the meeting and open the retained outputs after it. |
| 05 | Chat with WarpBot AI | Ask the assistant about the meeting they are in, or the documents their workspace holds. |

ROLE II OF IV

Host

The host owns the room from creation to record: **schedule, run, translate, record, publish.**

01	Create & schedule a meeting	Title, description, meeting type, participants, time and tags – including recurring series.
02	Configure the room	Set the room's language set, seat limit, approval and mute-on-entry behaviour.
03	Start / stop translation	Turn the live pipeline on and off; every run is a numbered translation session in the record.
04	Control the room	Invite, approve join requests, manage participants and start or stop recording.
05	Edit transcript & finalize	Correct any line while the transcript is open, then lock it so downstream outputs are stable.
06	Publish & share artifacts	Release the transcript, AI summary and recording under the workspace's retention policy.

ROLE III OF IV

Workspace Owner

This is the role that makes WarpTalk a **B2B product** rather than a meeting feature: money, people, policy and knowledge.

- | | | |
|----|----------------------|--|
| 01 | Create a workspace | Choose a plan, pay, and stand up the organization's own tenant with its slug and settings. |
| 02 | Billing & AI credits | Subscription, shared credit balance, usage per service and the invoice history behind it. |
| 03 | Members | Invite, approve and remove people; assign workspace roles and hand over ownership. |
| 04 | Policies | Language policy, membership rules, artifact retention and consent requirements. |
| 05 | Documents | Upload the organization's material so the assistant answers from its own knowledge. |
| 06 | Glossary | Pin the terms that must never be mistranslated – names, products, internal vocabulary. |

ROLE IV OF IV

WarpTalk Administrator

The platform side of the product – **one portal** across every workspace on the system.

- | | | |
|----|-------------------------------|---|
| 01 | Workspaces | Every tenant on the platform: status, plan, owner, members and lifecycle actions. |
| 02 | Users | Accounts across all workspaces, with suspension and role management. |
| 03 | Plans & billing | Define subscription plans, entitlements, credit economics and settle usage charges. |
| 04 | Platform policies | System-wide defaults that a workspace inherits when it has not overridden them. |
| 05 | Dashboard & audit | Platform usage, revenue and an audit trail of every privileged action taken. |
| 06 | Glossary & notices | Platform-level terminology and the notices pushed to every workspace. |

SECTION 06 OF 08

CORE WORKFLOWS

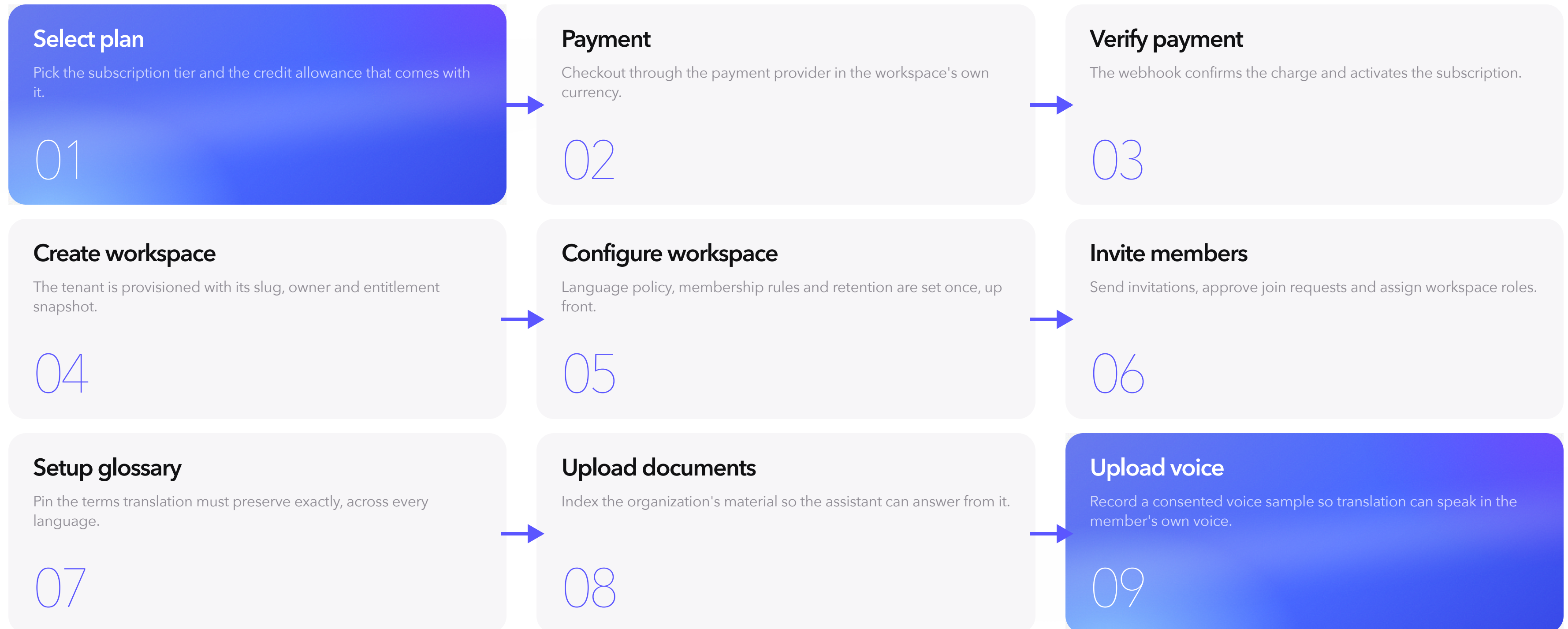
Primary flows across workspace and meeting operations.

06

WORKFLOW 01 OF 04

Setup workspace

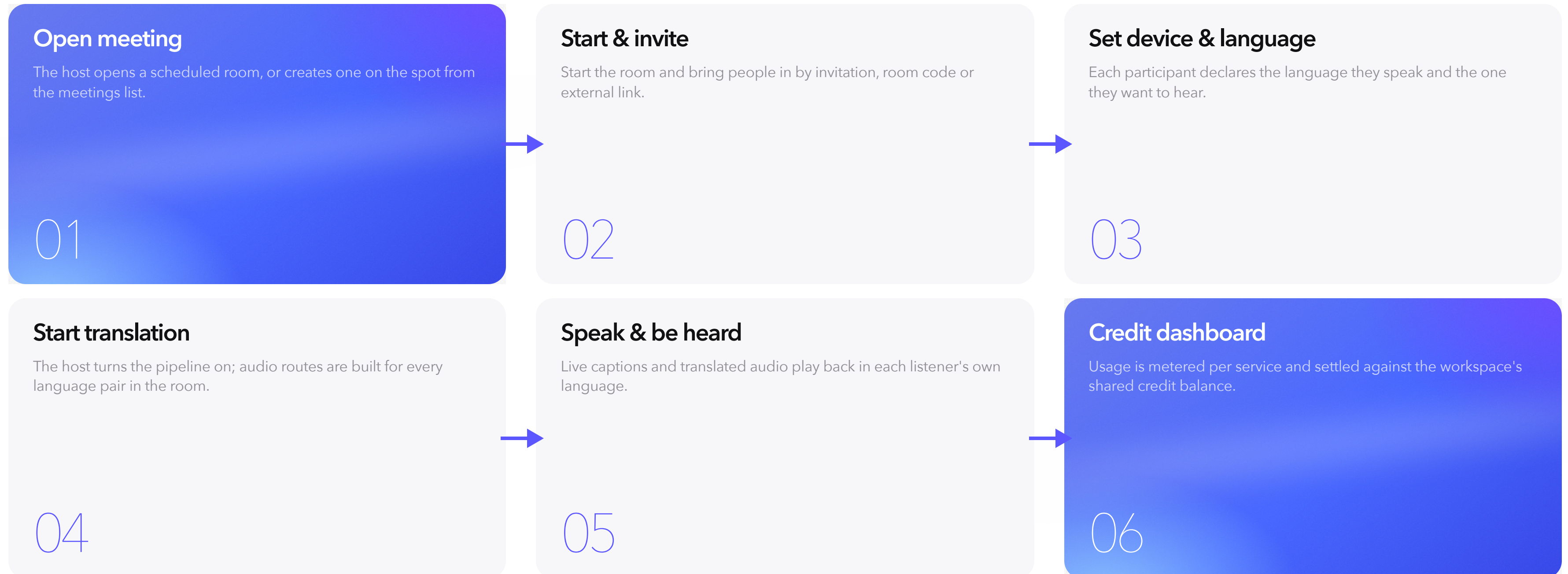
From a payment to a workspace that **already knows the organization** – its people, its words, its voices.



WORKFLOW 02 OF 04

Realtime translation meeting

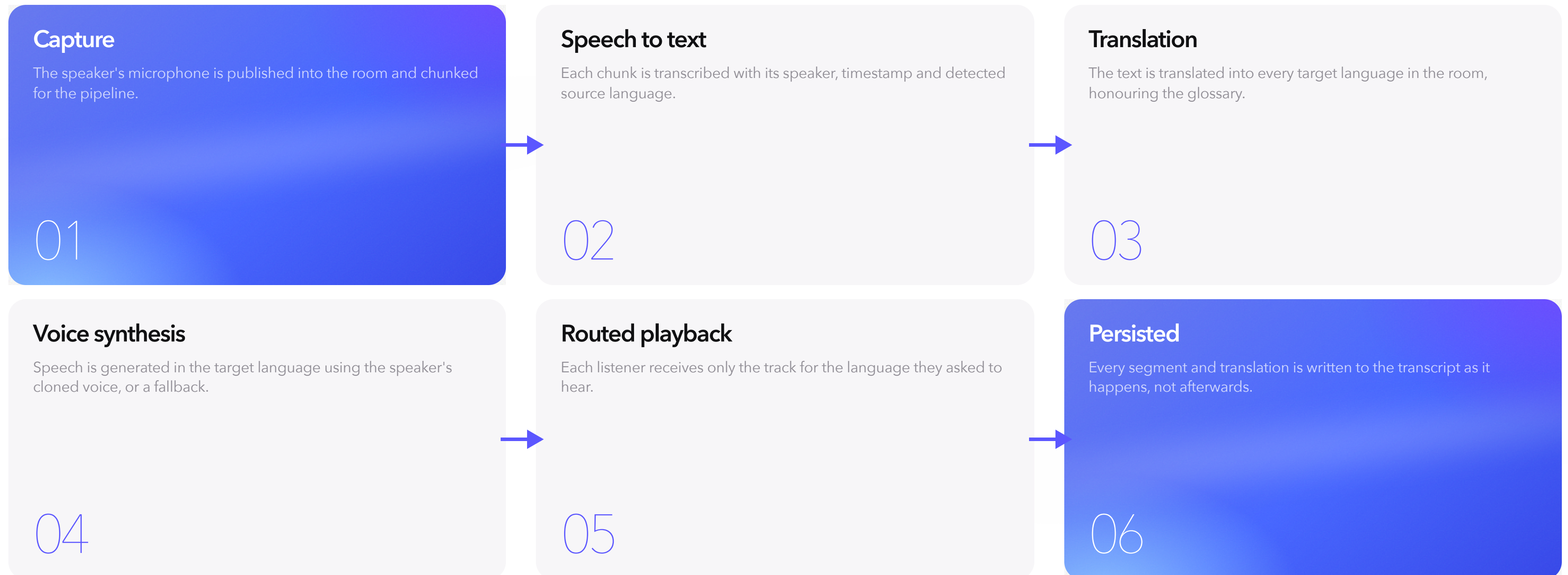
What the host does – from opening a room to watching the **credits it consumed**.



WORKFLOW 03 OF 04

Multilingual live meeting

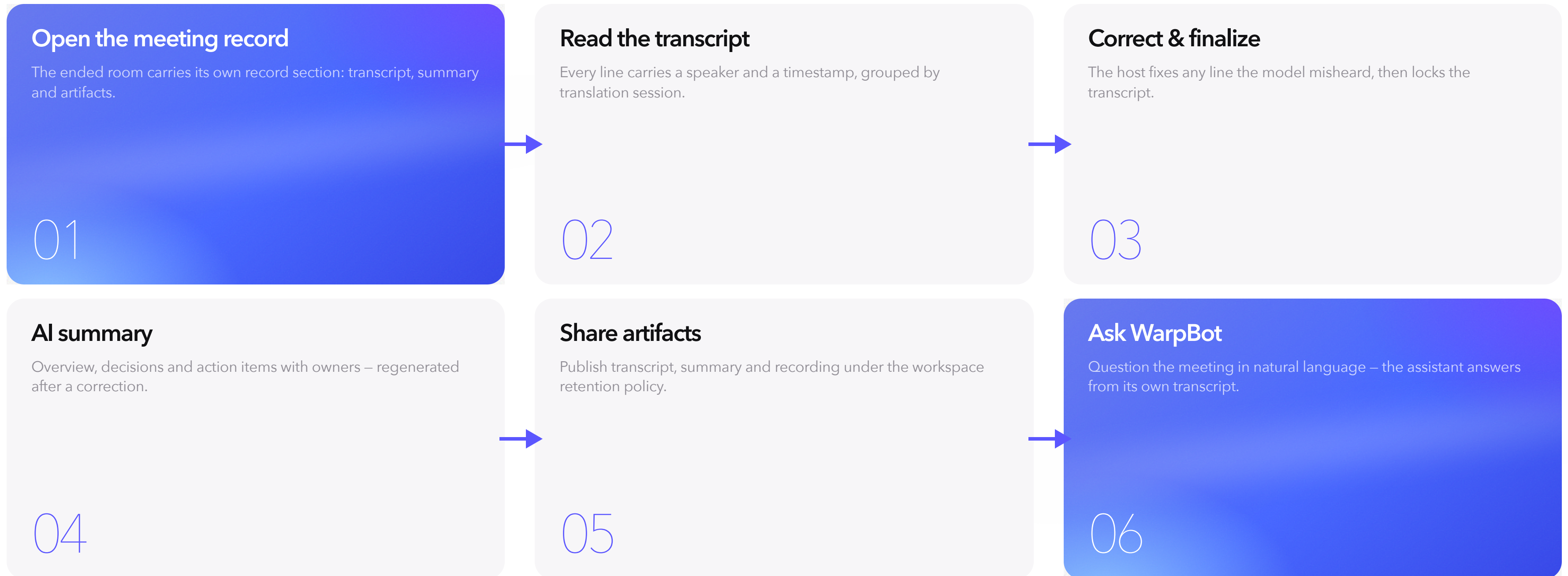
The same six seconds of speech, seen from inside the pipeline:
capture → **text** → **meaning** → **voice**.



WORKFLOW 04 OF 04

Review transcript & summary

The meeting does not end when the call does – it becomes a **record you can question**.



SECTION 07 OF 08

PROJECT ACHIEVEMENTS

What we did not do, what we achieved, and where it goes next.

07

SECTION VII · LIMITATIONS

What this release does not do

Stated before the results, so the results are read for what they are: **a scoped system**, not an unbounded claim.

01 A latency floor we cannot remove

Translated speech lands a few seconds behind the speaker: the audio has to be buffered before it can be transcribed at all.

02 A validated language set, not every language

The pipeline is proven on the languages we tested end to end. The picker is deliberately not a promise about the rest.

03 No translation-quality score

We show no confidence badge, because the only number we have measures speech recognition, not translation. Showing it would be a false signal.

04 One mixed recording, not one track per speaker

The room is captured as a single composite, so a recording cannot be cleanly re-transcribed afterwards.

05 Voice cloning needs consent and a clean sample

A clip below the quality bar is refused and the meeting falls back to a default voice rather than a bad imitation.

06 Web and desktop only

Participants join from a browser or the desktop app. There is no mobile client in this release.

SECTION VII · ACHIEVEMENTS

What we achieved

Four kinds of outcome – **a product, a pipeline, an organization model, and the evidence** that all three work.

OUTCOME 01

Product

- Built a B2B workspace-based AI meeting platform
- Supported multilingual meetings with voice and transcript
- Turned meetings into searchable records, summaries and action items

01

OUTCOME 02

AI & realtime

- Designed the speech → text → translation → voice pipeline
- Added voice-clone translation with consent and fallback
- Connected realtime audio, transcript stream and AI assistant flow

02

OUTCOME 03

Enterprise

- Added workspace roles, language policy and governance
- Added subscription, shared AI credits, usage tracking and audit
- Prepared admin and billing flows for organization-level operation

03

OUTCOME 04

Engineering evidence

- Requirements, design, test documents and user guides
- End-to-end demo flow across workspace, voice, meeting, record and admin
- Unit, integration, system and UAT testing evidence

04

SECTION VII · FUTURE WORK

Where this goes next

Every limit on the last slide is a **roadmap item**, and we already know which one we would take first.

NEXT 01

Sub-second delivery

Shrink the buffer so a translated sentence lands while the speaker is still in the room, not after the point has passed.

01

NEXT 02

Every language we can prove

Grow the validated set language by language, each one earning its place with measured quality.

02

NEXT 03

A track for every voice

Record each speaker separately, so any meeting can be re-transcribed, re-translated and re-dubbed later.

03

NEXT 04

Minutes a human can sign

A standard-format record with a named recorder on top of the AI summary – something an organization can file.

04

NEXT 05

Wherever the meeting already is

Extend the bridge past Google Meet to the other platforms teams live in.

05

NEXT 06

In your pocket

Bring the participant experience to mobile, alongside web and desktop.

06

SECTION 08 OF 08

QUESTIONS & ANSWERS

Discussion with the review committee.

08

END OF PRESENTATION

Thank you.

Speak once. **Be understood everywhere.**
We are ready for the committee's questions.

PROJECT

WarpTalk

TEAM

GSU26SE16

SUPERVISOR

Ms. Thân Thị Ngọc Vân

DEMO SCRIPT

Warptalk – Tab 8

SECTION VIII · DISCUSSION

Q&A

The floor is **yours**.
We are ready for the committee's questions.

LIVE DEMO

warptalk.vn

DEMO SCRIPT

Warptalk – Tab 8

TEAM

GSU26SE16

SUPERVISOR

Ms. Thân Thị Ngọc Vân